

Reading 5: The Behavioral Finance Perspective

- Expected utility (U) = Σ (U values of outcomes $\times$ Respective Prob)
- Subjective expected U of an individual = $\Sigma [u(x_i) \times \text{Prob}(x_i)]$
- Bayes' formula = $P(A|B) = [P(B|A) / P(B)] \times P(A)$
- Risk premium = Certainty equivalent – Expected value
- Perceived value of each outcome = $U = w(p_1) v(x_1) + w(p_2) v(x_2) + \dots + w(p_n) v(x_n)$
- Abnormal return (R) = Actual R – Expected R

Reading 8: Managing Individual Investor Portfolios

- After-tax (AT) *Real* required return (RR) %

$$= \frac{\text{Client's required expenditures in Year } n}{\text{Net Investable Assets}} = \frac{\text{Projected needs in Year } n}{\text{Net Investable Assets}}$$
- AT *Nominal* RR % = $\frac{\text{Projected needs in Year } n}{\text{Net Investable Assets}} + \text{Current Annual (Ann) Inflation (Inf) \%} = \text{AT real RR\%} + \text{Current Ann Inf\%}$ Or

$$\text{AT Nominal RR\%} = (1 + \text{AT Real RR\%}) \times (1 + \text{Current Ann Inf\%}) - 1$$

- Total Investable assets = Current Portfolio – Current year cash outflows + Current year cash inflows
- Pre-tax income needed = AT income needed / (1 – tax rate)
- Pre-tax Nominal RR = (Pre-tax income needed / Total investable assets) + Inf%

If Portfolio returns are tax-deferred:

- Pre-tax projected expenditure \$ = AT projected expenditure \$ / (1 – tax rate)
- Pre-tax real RR % = Pre-tax projected expenditures \$ / Total investable assets
- Pre-tax nominal RR = (1 + Pre-tax real RR %) $\times$ (1 + Inflation rate%) – 1

If Portfolio returns are NOT tax-deferred:

- AT real RR% = AT projected expenditures \$ / Total Investable assets
- AT nominal RR% = (1 + AT real RR%) $\times$ (1 + Inf%) – 1
- Procedure of converting nominal, pre-tax figures into real, after-tax return:
 - Real AT R = [Expected total R – (Expected total R of Tax-exempt Invest $\times$ wt of Tax-exempt Invest)] $\times$ (1 – tax

rate) + (Expected total R of Tax-exempt Invest $\times$ wt of Tax-exempt Invest) – Inf rate

Or

- Real AT R = [(Taxable R of asset class 1 $\times$ wt of asset class 1) + (Taxable R of asset class 2 $\times$ wt of asset class 2) + ... + (Taxable return of asset class n $\times$ wt of asset class n)] $\times$ (1 – tax rate) + (Expected total R of Tax-exempt Invest $\times$ wt of Tax-exempt Invest) – Infrate

Reading 9: Taxes and Private Wealth Management in a Global Context

- After-tax (AT) rate = Total taxes paid / Total taxable income
- AT R = $r \times (1 - t_i)$
- AT Future Accumulations after n years = $FVIF_i = \text{Initial Invest} \times [1 + r(1 - t_i)]^n$
- Tax drag (\$) on capital accumulation = Acc capital without tax – Acc capital with tax
- Tax drag (%) on capital accumulation = (Acc capital without tax – Acc capital with tax) / (Acc capital without tax – Initial investment)
- Returns-Based Taxes: Deferred Capital Gains:

- AT Future Accumulations after n years = $FVIF_{cg} = \text{Initial Invst.} \times [(1 + r)^n (1 - t_{cg}) + t_{cg}]$
 - Value of a capital gain tax deferral = AT future accumulations in deferred taxes – AT future accumulations in accrued annually taxes
7. Cost Basis
- Capital gain/loss = Selling price – Cost basis
 - AT Future Accumulation = $FVIF_{cgb} = \text{Initial Invst} \times [(1 + r)^n (1 - t_{cg}) + t_{cg} - (1 - B) t_{cg}] = \text{Initial Invst} \times [(1 + r)^n (1 - t_{cg}) + (t_{cg} \times B)]$
Where, B = Cost basis
 $t_{cg} \times B$ = Return of basis at the end of the Invst.horizon.
When cost basis = initial Invst $\rightarrow B=1$,
 $FVIF_{cg} = \text{Initial investment} \times [(1 + r)^n (1 - t_{cg}) + t_{cg}]$
8. Wealth-Based Taxes
- AT Future Acc = $FVIF_w = \text{Initial Invst} [(1 + r) (1 - t_w)]^n$
Where, t_w = Ann wealth tax rate
9. Blended Taxing Environments
- Proportion of total return from Dividends (p_d) which is taxed at a rate of t_d .
 $p_d = \text{Dividends (\$)} / \text{Total dollar return}$
 - Proportion of total return from Interest income (p_i) which is taxed at a rate of t_i .
 $p_i = \text{Interest (\$)} / \text{Total dollar return}$
 - Proportion of total return from Realized capital gain (p_{cg}) which is taxed at a rate of t_{cg} .
 $p_{cg} = \text{Realized Capital gain (\$)} / \text{Total dollar return}$
 - Unrealized capital gain return: Total Dollar Return = Dividends + Interest income + Realized Capital gain + Unrealized capital gain
Total realized tax rate = $[(p_i \times t_i) + (p_d \times t_d) + (p_{cg} \times t_{cg})]$
10. Effective Ann AT R = $r^* = r (1 - p_i t_i - p_d t_d - p_{cg} t_{cg}) = r (1 - \text{total realized tax rate})$
Where, r = Pre-tax overall return on the portfolio and r^* = Effective ann AT R
11. Effective Capital Gains Tax = $T^* = t_{cg} (1 - p_i - p_d - p_{cg}) / (1 - p_i t_i - p_d t_d - p_{cg} t_{cg})$
12. Future AT acc. = $FVIF_{\text{Taxable}} = \text{Initial Invst} [(1 + r^*)^n (1 - T^*) + T^* - (1 - B) t_{cg}]$
13. Initial Invst (1 + Accrual Equivalent R)ⁿ = Future AT Acc
14. Accrual Equivalent R = $(\text{Future AT Acc} / \text{Initial Invst})^{1/n} - 1$
15. Accrual Equivalent Tax Rates = $r (1 - T_{AE}) = R_{AE}$
 $T_{AE} = 1 - \frac{R_{AE}}{R}$
16. In Tax Deferred accounts (TDAs) Future AT Acc = $FVIF_{TDA} = \text{Initial Invst} [(1 + r)^n (1 - T_n)]$
17. In Tax-exempt accounts $FVIF_{\text{taxEx}} = \text{Initial Invst} (1 + r)^n$
- $FVIF_{TDA} = FVIF_{\text{taxEx}} (1 - T_n)$
18. AT asset wt of an asset class (%) = AT MV of asset class (\$) / Total AT value of Portfolio (\$)
19. AT Initial invst in tax-exempt accounts = $(1 - T_0)$
20. FV of a pretax \$ invested in a tax-exempt account = $(1 - T_0) (1 + r)^n$
21. FV of a pretax \$ invested in a TDA = $(1 + r)^n (1 - T_n)$
22. Investors AT risk = S.D of pre-tax R (1 – Tax rate) = $\sigma(1 - T)$
23. Tax alpha from tax-loss harvesting (or Tax savings) = Capital gain tax with unrealized losses – Capital gain tax with realized losses
Or
Tax alpha from tax-loss harvesting = Capital loss $\times$ Tax rate
24. Pretax R taxed as a short-term gain needed to generate the AT R equal to long-term AT R = Long-term gain after-tax return / (1 – short-term gains tax rate)

Reading 10: Estate Planning in a Global Context

1. Estate = Financial assets + Tangible personal assets + Immoveable property + Intellectual property
2. Discretionary wealth or Excess capital = Assets – Core capital
3. Core Capital (CC) Spending Needs =
$$\sum_{j=1}^N \frac{p(\text{Survival}_j) \times \text{Spending}_j}{(1+r)^j}$$
4. Expected Real spending = Real annual spending $\times$ Combined probability
5. CC needed to maintain given spending pattern = Annual Spending needs / Sustainable Spending rate
6. Tax-Free Gifts = $RV_{\text{TaxFreeGift}} = \frac{[1+r_g(1-t_{ig})]^n}{[1+r_e(1-t_{ie})]^n(1-T_e)}$
7. Relative value of the tax-free gift = $1 / (1 - T_e)$
8. Taxable Gifts = $RV_{\text{TaxableGift}} = \frac{[1+r_g(1-t_{ig})]^n(1-T_g)}{[1+r_e(1-t_{ie})]^n(1-T_e)}$

9. Value of a taxable gift (if gift & asset (bequeathed) have equal AT R) = $(1 - T_g) / (1 - T_e)$
10. The relative after-tax value of the gift when the donor pays gift tax and when the recipient's estate will not be taxable (assuming $r_g = r_e$ and $t_{ig} = t_{ie}$):

$$RV_{\text{TaxableGift}} = \frac{FV_{\text{Gift}}}{FV_{\text{Bequest}}} = \frac{[1+r_g(1-t_{ig})]^n(1-T_g+T_gT_e)}{[1+r_e(1-t_{ie})]^n(1-T_e)}$$

11. Size of the partial gift credit = Size of the gift $\times T_g T_e$
12. Relative value of generation skipping = $1 / (1 - T_1)$
13. Charitable Gratuitous Transfers =
$$RV_{\text{CharitableGift}} = \frac{FV_{\text{CharitableGift}}}{FV_{\text{Bequest}}} = \frac{(1+r_g)^n + T_a[1+r_e(1-t_{ie})]^n(1-T_e)}{[1+r_e(1-t_{ie})]^n(1-T_e)}$$
14. Credit method = $T_C = \text{Max} [T_R, T_S]$
15. Exemption method = $T_E = T_S$
16. Deduction method = $T_D = T_R + T_S - T_R T_S$

Reading 12: Lifetime Financial Advice: Human Capital, Asset Allocation, & Insurance

1. Total Wealth = Human Capital + Financial Wealth
2. Human capital =
$$HC(x) = \sum_{t=x+1}^n \frac{E[h_t]}{(1+r+v)^{t-x}}$$
3. Utility Function =
$$U = \frac{(W_{t+1} + H_{t+1})^{1-\gamma}}{1-\gamma} \rightarrow \text{For } \gamma \neq 1$$

$$U = \ln(W_{t+1} + H_{t+1}) \rightarrow \text{For } \gamma = 1$$
4. Optimization Function =
$$\max(\theta_x, \alpha_x) E[(1-D)(1 - \bar{q}_x)U_{\text{alive}}(W_{x+1} + H_{x+1}) + D(\bar{q}_x)U_{\text{dead}}(W_{x+1} + \theta_x)]$$

Reading 13: Managing Institutional Investor Portfolio

Defined-Benefit Plans:

1. Funded Status of Pension Plan (PP) = MV of PP assets – PV of PP liabilities
2. Min RR for a fully-funded PP = Discount rate used to calculate the PV of plan liabilities
3. Desired R for a fully-funded PP = Discount rate used to calculate the PV of plan liabilities + Excess Target return

4. Net cash outflow = Benefit payments – Pension contributions

Foundations

5. Min R requirement (req) = Min Ann spending rate + InvstMgmtExp + Expected Inf rate
Or
Min Rreq = $[(1 + \text{Min Ann spending rate}) \times (1 + \text{Invst Mgmt. Exp}) \times (1 + \text{Expected Inf rate})] - 1$

6. Foundation's liquidity req = Anticipated cash needs (captured in a foundation's distributions prescribed by minimum spending rate*) + Unanticipated cash needs (not captured in a foundation's distributions prescribed) – Contributions made to the foundation.

* It includes Minimum annual spending rate (including "overhead" expenses e.g. salaries) + Investment management expenses

Endowments

7. Ann Spending (\$) = % of an endowment's current MV
Or
AnnSpending (\$) = % of an endowment's avg trailing MV
8. Simple spending rule = $\text{Spending}_t = \text{Spending rate} \times \text{Endowment's End MV}_{t-1}$

9. Rolling 3-yr Avg spending rule = $\text{Spending}_t = \text{Spending rate} \times \text{Endowment's Avg MV of the last 3 fiscal yr-ends i.e.}$

→ $\text{Spending}_t = \text{Spending rate} \times (1/3) [\text{Endowment's End MV}_{t-1} + \text{Endowment's End MV}_{t-2} + \text{Endowment's End MV}_{t-3}]$

10. Geometric smoothing rule = $\text{Spending}_t = \text{WghtAvg of the prior yr's spending adjusted for Inf} + \text{Spending rate} \times \text{Beg MV of the prior fiscal yr i.e.}$

→ $\text{Spending}_t = \text{Smoothing rate} \times [\text{Spending}_{t-1} \times (1 + \text{Inf}_{t-1})] + (1 - \text{Smoothing rate}) \times (\text{Spending rate} \times \text{Beg MV}_{t-1} \text{ of the endowment})$

11. Min ReqRoR = Spending rate + Cost of generating Invst R + Expected Infrate
Or
Min ReqRoR = $[(1 + \text{Spending rate}) \times (1 + \text{Cost of generating Invst R}) \times (1 + \text{Expected Inf rate})] - 1$

12. Liquidity needs = Ann spending needs + Capital commitments + Portfolio rebalancing expenses – Contributions by donor

13. Neutrality Spending Rate = Real expected R = Expected total R – Inf

Life Insurance Companies

14. Cash value = Initial premium paid + Any accrued interest on that premium

15. Policy reserve = PV of future benefits - PV of future net premiums

16. Surplus = Total assets of an insurance company - Total liabilities of an insurance company

Non-Life Insurance Companies

17. Combined Ratio = $(\text{Total amount of claims paid out} + \text{Insurer's operating costs}) / \text{Premium income}$

Banks

18. Net interest margin =
$$\frac{(\text{Interest Income} - \text{Interest Expense})}{\frac{\text{Avg Earning Assets}}{\text{Net Interest Income}}} = \frac{\text{Net Interest Income}}{\text{Avg Earning Assets}}$$

19. Interest spread = Avg yield on earning assets – Average percent cost of interest-bearing liabilities

20. Leverage-adjusted duration gap (LADG) = $D_A - (k \times D_L)$
Where, $k = \text{MV of liabilities} / \text{MV of assets} = L/A$

21. Change in MV of net worth of a bank (resulting from interest rate shock) $\approx$ - LADG $\times$ Size of bank $\times$ Size of interest rate shock

Reading 14: Linking Pension Liabilities to Assets

1. Value of liability = $V_L = \sum_t \frac{B_t}{(1+r_t)^t}$
where, B_t = Benefit payments at time t
2. Value of an asset = $V_B = \sum_t \frac{CF_t}{(1+r_t)^t}$
3. Intrinsic value of Future wage liability =

$$V_{L-FW} = \frac{B}{r-g} \times \frac{((1+g)^s - 1) \times ((1+r)^{d-s} - 1)}{(1+r)^d}$$

where, s = yrs till retirement
 d = yrs till demise and subsequent termination of the obligation

Reading 15: Capital Market Expectations

1. Precision of the estimate of the population mean $\approx 1 / \sqrt{\text{no of obsvs}}$
2. Multiple-regression analysis: $A = \beta_0 + \beta_1 B + \beta_2 C + \varepsilon$
3. Time series analysis: $A = \beta_0 + \beta_1 \text{ Lagged values of } A + \beta_2 \text{ Lagged values of } B + \beta_2 \text{ Lagged values of } C + \varepsilon$
4. Shrinkage Estimator = (Wt of historical estimate $\times$ Historical parameter estimate) + (Wt of Target parameter estimate $\times$ Target parameter estimate)

5. Shrinkage estimator of Cov matrix = (Wt of historical Cov $\times$ Historical Cov) + (Wt of Target Cov $\times$ Target Cov)

6. Vol in Period $t = \sigma_t^2 = \beta \sigma_{t-1}^2 + (1 - \beta) \varepsilon_t^2$
7. Multifactor Model: R on Asset $i = R_i = a_i + b_{i1}F_1 + b_{i2}F_2 + \dots + b_{iK}F_K + \varepsilon_i$

8. Value of asset at time t_0

$$= \sum_{t=1}^{\infty} \frac{CF \text{ at time } t}{(1 + \text{Discount rate})^t}$$
9. Expected RoR on Equity =

$$\frac{\text{Div per share at time } 0 (1 + \text{LT g rate})}{\text{Current share price}} + \text{LT g rate}$$

$$= \text{Div Yield} + \text{Capital Gains Yield}$$

10. Nominal GDP = Real g rate in GDP + Expected long-run Inf rate

11. Earnings g rate = Nominal GDP g rate + Excess Corp g (for the index companies)

12. Expected RoR on Equity $\approx \frac{D}{P} - \Delta S + i + g + \Delta PE$
 $-\Delta S$ = Positive repurchase yield
 $+\Delta S$ = Negative repurchase yield
 ΔPE = Expected Repricing Return
13. Labor supply g = Pop g rate + Labor force participation g rate

14. Expected income R = $D/P - \Delta S$

15. Expected nominal earnings g R = $i + g$

16. Expected Capital gains R = Expected nominal earnings grate + Expected repricing R

17. Asset's expected return $E(R_i) = R_f + (RP)_1 + (RP)_2 + \dots + (RP)_K$

18. Expected bond R $[E(R_b)] = \text{Real } R_f + \text{Inf premium} + \text{Default RP} + \text{Illiquidity P} + \text{Maturity P} + \text{Tax P}$

19. Inf P = Avg Inf rate expected over the maturity of the debt + P (or discount) for the prob attached to higher Inf than expected (or greater disinflation)

20. Inf P = Yield of conventional Govt. bonds (at a given maturity) – Yield on Inf-indexed bonds of the same maturity

21. Default RP = Expected default loss in yield terms + P for the non-diversifiable risk of default

22. Maturity P = Interest rate on longer-maturity, liquid Treasury debt - Interest rate on short-term Treasury debt

23. Equity RP = Expected ROE (e.g. expected return on the S&P 500) – YTM on a long-term Govt. bond (e.g. 10-year U.S. Treasury bond R)

24. Expected ROE using Bond-yield-plus-RP method = YTM on a LT Govt bond + Equity RP

25. Expected ROA $E(R_i) = \text{Domestic } R_f + R + (\beta_i) \times [\text{Expected } R \text{ on the world market portfolio} - \text{Domestic } R_f \text{ rate of } R]$

Where, β_i = The asset's sensitivity to R on the world mktportf = $\text{Cov}(R_i, R_M) / \text{Var}(R_M)$

26. Asset class RP_i = Sharpe ratio of the world market portfolio $\times$ Asset's own volatility (σ_i) $\times$ Asset class's correlation with the world mktportf ($\rho_{i,M}$)

$$RP_i = (RP_M / \sigma_M) \times \sigma_i \times \rho_{i,M}$$

Where, Sharpe Ratio of the world market portfolio = Expected excess R / S.D of the world mktportf $\rightarrow$ represents systematic or non-diversifiable risk = RP_M / σ_M

27. RP for a completely segmented market (RP_i) = Asset's own volatility (σ_i) $\times$ Sharpe ratio of the world mktportf
28. RP of the asset class, assuming partial segmentation = (Degree of integration $\times$ RP under perfectly integrated markets) + ($\{1 - \text{Degree of integration}\} \times RP$ under completely segmented markets)
29. Illiquidity P = Required RoR on an illiquid asset at which its Sharpe ratio = mkt's Sharpe ratio – ICAPM required RoR
30. Cov b/w any two assets = Asset 1 beta $\times$ Asset 2 beta $\times$ Var of the mkt

$$31. \text{Beta of asset 1} = \left(\frac{\sigma_1 \times \rho(1, m)}{\sigma_m} \right)$$

$$32. \text{Beta of asset 2} = \left(\frac{\sigma_2 \times \rho(2, m)}{\sigma_m} \right)$$

33. GDP (using expenditure approach) = Consumption + Invst + Δ in Inventories + Govt spending + (Expo- Impo)

34. Output Gap = Potential value of GDP – Actual value of GDP

35. Neutral Level of Interest Rate = Target Inf Rate + Eco g

36. Taylor rule equation: $R_{\text{optimal}} = R_{\text{neutral}} + [0.5 \times (\text{GDP}_{\text{forecast}} - \text{GDP}_{\text{trend}})] + [0.5 \times (I_{\text{forecast}} - I_{\text{target}})]$

37. Trend g in GDP = g from labor inputs + g from Δ in labor productivity

38. g from labor inputs = g in potential labor force size + g in actual labor force participation

39. g from Δ in labor productivity = g from capital inputs + TFP g^*
- TFP g = g associated with increased efficiency in using capital inputs.

40. GDP $g = \alpha + \beta_1 \text{Consumer spending } g + \beta_2 \text{Investment } g$

41. Consumer spending $g = \alpha + \beta_1 \text{Lagged consumer income } g + \beta_2 \text{Interest rate}$

42. Investment $g = \alpha + \beta_1 \text{Lagged GDP } g + \beta_2 \text{Interest rate}$

43. Consumer Income g = Consumer spending growth lagged one period

Reading 16: Equity Market Valuation

1. Cobb-Douglas Production Function $Y = A \times K^\alpha \times L^\beta$

Where, Y = Total real economic output

A = Total factor productivity (TFP)

K = capital stock

α = Output elasticity of K

L = Labor input

β = Output elasticity of L

2. Cobb-Douglas Production Function Y (assuming constant R to Scale) = $\ln(Y) = \ln(A) + \alpha \ln(K) + (1 - \alpha) \ln(L)$

Or

$$\frac{\Delta Y}{Y} \approx \frac{\Delta A}{A} + \alpha \frac{\Delta K}{K} + (1 - \alpha) \frac{\Delta L}{L}$$

3. Solow Residual = $\% \Delta TFP = \% \Delta Y - \alpha (\% \Delta K) - (1 - \alpha) \% \Delta L$

4. H-Model: Value per share at time 0 =
- $$\frac{D_0}{\text{Discount rate} - \text{LT sustainable Div g rate}} \times \left[(1 + \text{LT sustainable Div g rate}) + \frac{\text{Super normal g period}}{2} \times (\text{ST higher Div g rate} - \text{LT sustainable Div g rate}) \right]$$
5. Gordon g Div discount model: Value per share at time 0 = $\frac{(D_0) \times (1+g)}{r-g}$
6. Forward justified P/E = $\frac{\text{Intrinsic value}}{\text{Yr ahead expected Earnings}}$
7. Fed Model:
- $$\frac{\text{Fwd Operating Earnings (E}_1\text{)}}{\text{Index Level (P}_0\text{)}} = \text{Long-term US Treasury securities}$$
8. Yardeni Model: $= \frac{E_1}{P_0} = y_B - d \times LTEG$

Where, E_1/P_0 = Justified (forward) earnings yield on equities
 y_B = Moody's A-rated corporate bond yield
 LTEG = Consensus 5-yr earnings g forecast for the S&P 500
 d = Discount or Weighting factor that represents the weight assigned by the market to the earnings projections

9. Yardeni estimated fair value of P/E ratio =
- $$\frac{P_0}{E_1} = \frac{1}{y_B - d \times LTEG}$$

10. Fair value of equity mkt under Yardeni Model (P_0) = $P_0 = \frac{E_1}{y_B - d \times LTEG}$

11. Discount/weighting factor (d) =

$$d = \frac{y_B - \frac{E_1}{P_0}}{LTEG}$$

12. 10-year Moving Average Price/Earnings [P / 10-year MA (E)] =
- $$\frac{\text{Real (or Inf-adjusted) S\&P 500 Price Index}}{\text{Moving Avg of preceding 10 yrs of Real or Inf adj Earnings}}$$

*The stock index and reported earnings are adjusted for Inflation using the CPI

13. Real Stock Price Index $_t$ = (Nominal SPI $_t$ × CPI $_{\text{base yr}}$) / CPI $_t$
14. Real Earnings $_t$ = (Nominal Earnings $_t$ × CPI $_{\text{base year}}$) / CPI $_{t+1}$

15. Tobin's q = $\frac{\text{MV of debt} + \text{MV of equity}}{\text{Replacement cost of assets}}$
- $$\text{Equity q} = \frac{\text{Equity Mkt Cap}}{\text{Net Worth}} = \frac{\text{Price per share} \times \text{No of Shares O/S}}{\text{Replacement cost of assets} - \text{MV of liabilities}}$$

Reading 17: Asset Allocation

- Req R = [(1 + Spending rate) × (1 + Expected Inf %) × (1 + Cost of earning Invst R)] – 1
- Risk-adj Expected R = Expected return for mix 'm'* – (0.005 × Investor's risk aversion × Var of R for mix 'm'*)
- Risk Penalty = 0.005 × Investor's risk aversion × Var of R for mix 'm'*
*expressed as % rather than as decimals
- Safety First Ratio = $\frac{\text{Expected Portfolio R} - \text{Threshold level}}{\text{Portfolio S.D.}}$
- Include asset in the portfolio when:
 $\frac{E(R_{\text{new}}) - R_F}{\sigma_{\text{new}}} > \left[\frac{E(R_{\text{new}}) - R_F}{\sigma_{\text{new}}} \right] \text{Corr}(R_{\text{new}}, R_p)$
- Contribution of Currency risk = Vol of asset R in domestic ¢ – Vol of asset R in local ¢
Where Vol = volatility
- Funding Ratio = $\frac{\text{Market Value of Pension Assets}}{\text{Present Value of Pension Liabilities}}$
- $U_m^{ALM} = E(SR_m) - 0.005 R_A \sigma^2(SR_m)$
• U_m^{ALM} = Surplus objective function's expected value for a particular asset mix m, for a particular investor with the specified risk aversion.

- $E(SR_m) = \text{Expected surplus return for asset mix 'm'} = \frac{\Delta \text{ in asset value} - \Delta \text{ in liability value}}{\text{Initial Asset Value}}$
 - $\sigma^2(SR_m) = \text{Var of the surplus R for the asset mix m in \%}$
 - $R_A = \text{Risk-aversion level}$
9. Human Capital (t)

$$= \sum_{j=t}^T \frac{\text{Expected Earnings at age j}}{(1 + \text{discount rate})^{j-t}}$$
 t = current age T = life expectancy

Reading 18: Currency Management: An Introduction

1. Bid Fwd rate = Bid Spot exchange (X) rate

$$+ \frac{\text{Bid Fwd points}}{10,000}$$
2. Offer Fwd rate = Offer Spot X rate +

$$\frac{\text{Offer Fwd points}}{10,000}$$
3. $\text{FwdPrem/Disc \%} = \frac{\text{spot X rate} - \left(\frac{\text{fwd pnts}}{10,000}\right)}{\text{spot X rate}} - 1$
4. To convert spot rate into a forward quote when points are represented as %,

$$\text{Spot X rate} \times (1 + \% \text{ prem})$$

$$\text{Spot X rate} \times (1 - \% \text{ disc})$$
5. Mark-to-MV on dealer's position =

$$\frac{\text{Settlement day CF}}{1 + \text{Disc rate} \times \left(\frac{t}{T}\right)}$$

6. CF at settlement = Original contract size $\times$ (All-in-fwd rate for new, offsetting fwd position – Original fwd rate)

7. Hedge Ratio =

$$\frac{\text{Nominal Value of derivatives contract}}{\text{MV of the hedged asset}}$$

8. $R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1$

9. $R_{DC} \text{ (for multiple foreign assets)} =$

$$\sum_{i=1}^n \omega_i (1 + R_{FC,i})(1 + R_{FX,i}) - 1$$

10. Total risk of DC returns =

$$= \sigma^2(R_{DC}) \approx \sqrt{\frac{\sigma^2(R_{FC}) + \sigma^2(R_{FX}) + [2\sigma(R_{FC})\sigma(R_{FX})\rho(R_{FC}, R_{FX})]}{}}$$

11. $\% \Delta \text{ in spot X rate } (\% \Delta S_{H/L}) = \text{Interest rate on high-yield currency } (i_H) - \text{Interest rate on low-yield currency } (i_L)$

12. Forward Rate Bias = $\frac{F_{P/B} - S_{P/B}}{S_{P/B}} =$

$$\frac{(i_P - i_B) \left(\frac{t}{360}\right)}{1 + i_B \left(\frac{t}{360}\right)}$$

13. Net delta of the combined position =
 Option delta + Delta hedge

14. Size of Delta hedge (that would set net delta of the overall position to 0) =
 Option's delta $\times$ Nominal size of the contract

15. Long Straddle = Long atm put opt (with delta of -0.5) + Long atm call opt (with delta of +0.5)

16. Short Straddle = Short ATM put opt (with delta of -0.5) + Short ATM call opt (with delta of +0.5)
 ATM = at the money
 opt = option

17. Long Strangle: Long OTM put option + Long OTM call opt
 OTM = out of the money

18. Long Risk reversal = Long Call opt + Short Put opt

19. Short Risk reversal = Long Put opt + Short Call opt

20. Short seagull position = Long protective (ATM) put + Short deep OTM Call opt + Short deep OTM Put opt

21. Long seagull position = Short ATM call + Long deep-OTM Call opt + Long deep-OTM Put opt

22. Hedge ratio =

$$\frac{\text{Principal face value of the derivatives contract used as a hedge}}{\text{Principal face of the hedged asset}}$$

$$23. \text{ Min or Optimal hedge ratio} = \rho(R_{DC}, R_{FX}) \times \left[\frac{S.D(R_{DC})}{S.D(R_{FX})} \right]$$

Reading 19: Market Indexes and Benchmarks

1. Periodic R (Factor model based) = $R_p = a_p + b_1F_1 + b_2F_2 + \dots + b_KF_K + \varepsilon_p$
2. For one factor model $R_p = a_p + \beta_p R_1 + \varepsilon_p$
Where, R_1 = periodic R on mktindex
 a_p = "zero factor"
 β_p = beta = sensitivity
 ε_p = residual return
3. MV of stock = No of Shares Outstanding $\times$ Current Stock Mkt Price
4. Stock wgt(float-weighted index) = Mkt-cap wgt $\times$ Free-float adjustment factor
5. Price-weighted index (PWI) = $(P_1 + P_2 + \dots + P_n) / n$

Reading 20: Fixed-Income Portfolio Management – Part I

1. Steps to calculate PVDistribution (PVD) of CFs:
a) Wgt of Index's total MV attributable to CFs in each period = $\frac{\text{PV of CFs from B index for specific period}}{\text{PV of Total CFs from B}}$
where B = Benchmark

- b) Contribution of each period's CFs to portfolio D = D of each period $\times$ Wgt of index CFs in specific period

$$c) \text{ Benchmark's PVD} = \frac{\text{Cont of each period's CFs to portfolio D}}{\text{sum of all the periods' D cont}}$$

2. Active R = Portfolio's R – B Index's R
3. Tracking Risk = S.D of Active R = $\left(\frac{\sum (\text{Active R} - \text{Mean Active R})^2}{n-1} \right)^{\frac{1}{2}}$
4. Semi-annual Total R = $\left(\frac{\text{Total Future Dollars}}{\text{Full Price of the Bond}} \right)^{\frac{1}{n}} - 1$
5. Dollar D = D $\times$ Portfolio Value $\times$ 0.01
6. Portfolio's Dollar D = Sum of dollar D of securities in portfolio
7. Rebalancing Ratio = $\frac{\text{Original Dollar D}}{\text{New Dollar D}}$
8. Cash required for rebalancing = $(\text{Rebalancing ratio} - 1) \times (\text{total new MV of portfolio})$
9. Controlling Position = Target Dollar D – Current Dollar D
10. Contribution of bond/sector to the portfolio D = $\left(\frac{\text{MV of bond or sector in the Portfolio}}{\text{Total Portfolio Value}} \right) \times \text{Effective D of bond or sector}$

11. Spread D of a Portfolio = Market wgtavg of the sector spread D of the individual securities

$$12. \text{ Net safety rate of return (Cushion Spread)} = \text{Immunized Rate} - \text{Min acceptable R}$$

13. Dollar safety margin = Current bond portfolio value - PV of the required terminal value at new interest rate
14. Economic Surplus = MV of assets – PV of liabilities

$$15. \text{ Confidence Interval} = \text{Target Return} \pm (k) \times (\text{S.D of Target R})$$

where, k = number of S.D around the expected target R

Reading 22: Fixed-Income Portfolio Management – Part II

1. D of Equity = $\frac{(\text{D of Assets} \times \text{Assets}) - (\text{D of Liab} \times \text{Liab})}{\text{Equity}}$
2. R_p = Portfolio RoR = $\frac{\text{Profit on borrowed funds} + \text{Profit on Equity}}{\text{Amount of Equity}} = [B \times (r_F - k) + E \times r_F] / E$
 $= r_F + \left[\frac{B}{E} \times (r_F - k) \right]$
3. Dollar interest = $\frac{\text{Amount borrowed} \times \text{Repo rate} \times \text{Repo term}}{360}$

4. New bond MV = $\frac{\text{Dollar D of Old Bond}}{\text{Duration of New Bond}} \times 100$
5. New bond Par value = $\frac{\text{Dollar D of Old Bond}}{\text{New Dollar D per Bond}} \times 100$
6. Shortfall risk = $\frac{\text{No of obs below the Target R}}{\text{Total No of Observations}}$
7. Target dollar D = Current dollar D without futures + Dollar D of futures position
8. No of Futures Contracts = $\frac{\text{Target \$ D} - \text{Current \$ D without futures}}{\text{\$ D per futures contract}}$
9. Dollar duration of futures contract = $\frac{\text{\$ D of Cheapest to Deliver issue}}{\text{Factor exposure of Hedging instrument}} \times \text{CF for CTD Issue}$
10. Hedge Ratio = $\frac{\text{Factor exposure of the bond (portfolio) to be hedged}}{\text{Factor exposure of Hedging instrument}}$
or
Hedge Ratio = $\left[\frac{\text{Duration of the bond to be hedged} \times \frac{\text{Price of the bond to be hedged}}{\text{Duration of the CTD bond} \times \frac{\text{Price of the CTD}}{\text{Price of the CTD}}}}{\text{Price of the CTD}} \right] \times$
(Conversion factor for CTD bond)
11. Basis = Cash (spot) price – Futures price
12. Yield on bond to be hedged = $a + (\text{Yield Beta} \times \text{yield on CTD Issue}) + \text{Error}$
13. Hedge ratio = $\frac{D_H - P_H}{D_{\text{CTD}} P_{\text{CTD}}} \times \text{Conversion factor for CTD Issue} \times \text{Yield Beta}$
14. Interest rate Swap (fixed-rate receiver/floating rate payer) = Long a fixed-rate bond + Short a floating-rate bond
15. \$ D of a swap for a fixed-rate receiver (floating rate payer) = \$ D of a fixed-rate bond – \$ D of a floating-rate bond
OR
\$ D of a swap for a fixed-rate receiver $\approx$ \$ D of a fixed-rate bond
16. Interest Rate Swap (fixed-rate payer/floating rate receiver) = Long a floating-rate bond + short a fixed-rate bond
17. \$ D of a swap for a fixed-rate payer = \$ D of a floating-rate bond – \$ D of a fixed-rate bond
OR
\$ D of a swap for a fixed-rate payer $\approx$ –\$ D of a fixed-rate bond
18. \$ D of a portfolio that includes a swap = \$ D of assets – \$ D of liabilities + \$ D of a swap position
19. D for an Option = $\Delta \text{ of Option} \times \text{D of Underlying Instrument} \times (\text{Price of underlying}) / (\text{price of Opt instrument})$
where Opt = Option
20. Payout to Opt Buyer or Opt value = $\text{MAX} [(\text{Strike value} - \text{Value at maturity}), 0]$
21. Credit spread call Opt value/Payoff = $\text{Max} [(\text{Spread at the opt maturity} - \text{Strike spread}) \times \text{NP} \times \text{Risk factor}, 0]$
22. Credit Forward Payoff = $(\text{Credit spread at the forward contract at maturity} - \text{Contracted credit spread}) \times \text{NP} \times \text{Risk factor}$
23. Change in Foreign bond Value (In terms of change in foreign yield only) = $\text{Duration} \times \Delta \text{ Foreign yield} \times 100$
24. Change in Foreign bond Value (when domestic rates change) = $\text{Duration} \times \text{Yield beta} \times \Delta \text{ Domestic yield} \times 100$
25. $\Delta \text{ Yield}_{\text{Foreign}} = \alpha + \text{Yield beta or country beta } (\beta) (\Delta \text{ yield}_{\text{Domestic}}) + \varepsilon$
26. Estimated % $\Delta \text{ Value}_{\text{Foreign}} = \text{Yield beta} \times \Delta \text{ Domestic yield}$
27. D Cont of Domestic Bond = $\text{Wght of domestic bond in Portfolio} \times \text{D of Domestic Bond}$
28. D Cont of Foreign Bond = $\text{Wght of foreign bond in Portfolio} \times \text{D of Foreign Bond} \times \text{Country beta}$

29. Portfolio D = D Cont of Domestic Bond +
D Cont of Foreign Bond

30. Interest rate parity = $F = S_0 \times [(1 + i_D) \div (1 + i_F)]$

31. Forward Premium = $(F - S_0) / S_0 = (i_D - i_F) / (1 + i_F)$

32. Forward Premium (as first order linear approx) = $(F - S_0) / S_0 \approx i_D - i_F$

33. Unhedged R = Foreign bond R in local curr + curr return (or FC appreciation)

34. Hedged R (HR) = Foreign bond R in local curr + Forward discount (premium)

35. HR = Domestic risk-free rate + bond's local premium = $HR = i_d + (r_1 - i_f)$

36. Bond's local risk premium = Bond's return - local risk-free rate $(r_1 - i_f)$

37. Breakeven Spread change = $\frac{\% \Delta \text{Price}}{D} \times 100$

Reading 23: Equity Portfolio Management

1. Active R = Portf's R - B's return
B = benchmark
2. Tracking Risk (active risk) = ann S.D of active R

3. Information Ratio =
$$\frac{\text{Active R}}{\text{Tracking Risk or Active Risk}}$$

4. Passive investment using Equity Index futures = Long cash + Long futures on the underlying index

5. Passive investment using Equity total return swaps = Long cash + Long swap on the index

6. R on Portf = $b_0 + (b_1 \times R \text{ on Index style 1}) + (b_2 \times R \text{ on Index style 2}) + \dots + (b_n \times R \text{ on Index style n}) + \varepsilon$

7. RoR of Equitized Mkt neutral strategy = (G/L on long & short securities positions + G/L on long futures position + Interest earned on cash from short sale) / Portfolio Equity

8. Active wgt = Stock's wgt in actively managed portf - Stock's wgt in B

9. Info Ratio $\approx$ Info Coefficient $\times$
 $\sqrt{\text{Info Breadth}}$

10. Risk-adjusted Expected Active R =
 $U_A = r_A - \lambda_A \times \sigma_A^2$

11. Portfolio Active R =
$$\sum_{i=1}^n \text{Wgt assigned to ith Mngr (hAi)} \times \text{Active R of the ith Mngr (rAi)}$$

12. Portfolio Active Risk =

$$\sqrt{\sum_{i=1}^n \frac{\text{Wgt assigned to ith Mngr}^2 \times \text{Active R of ith Manager}^2}{}}$$

13. Mngr's "true" active R = Mngr's R - Mngr's Normal B

14. Mngr's "misfit" active R = Mngr's normal B R - Investor's B

15. Total Active Risk =

$$\sqrt{(\text{True Active Risk})^2 + (\text{Misfit Active Risk})^2}$$

Where,

True active risk = S.D of true active R

Misfit risk = S.D of misfit active R

16. True Information Ratio =

$$\frac{\text{Mngr's True Active R}}{\text{Mngr's True Active risk}}$$

17. Investors' net of fees alpha = Gross of fees alpha (or mngr's alpha) - Investment mgmt fees

Reading 24: Alternative Investments Portfolio Management

1. Minority interest discount (\$) = marketable controlling interest value (\$) $\times$ minority interest(%) discount = (investor's interest in equity $\times$ total equity value) $\times$ minority interest discount(%)

2. Marketable minority interest (\$) =
Marketable controlling interest value (\$) –
minority interest discount (\$)
3. Marketability discount (\$) = Marketable
minority interest (\$) × marketability
discount (%)
4. Non-Marketable minority interest (\$) =
Marketable minority interest (\$) -
marketability discount (\$)
5. Total R on Commodity Index = Collateral
R + Roll R + Spot R
6. Monthly Roll R = Δ in futures contract
price over the month - Δ in spot price over
the month
7. Compensation structure of Hedge Funds
(comprises of) Management fee (or AUM
fee) + Incentive fee
8. Management fee = % of NAV (net asset
value generally ranges from 1-2%)
9. Incentive fee = % of profits (specified by
the investment terms)
10. Incentive fee (when High Water mark
Provision) = (positive difference between
ending NAV and HWM NAV) × incentive
fee %.
11. Hedge Fund R = [(End value) – (Beg
value)] / (Beg value)

$$12. \text{Rolling R} = \text{RR}_{n,t} = (R_t + R_{t-1} + R_{t-2} + \dots + R_{t-(n-1)}) / n$$

$$13. \text{Downside Deviation} = \sqrt{\frac{\sum_{t=1}^n [\min(r_t - r^*, 0)]^2}{n-1}}$$

where, $r^* = \text{threshold}$

$$14. \text{Semi-deviation} = \sqrt{\frac{\sum_{t=1}^n [\min(r_t - \text{avg. monthly return}, 0)]^2}{n-1}}$$

$$15. \text{Sharpe ratio} = (\text{Annualized RoR} - \text{Annualized Rf rate}) / \text{Annualized S.D.}$$

$$16. \text{Sortino Ratio} = (\text{Annualized RoR} - \text{Annualized Rf}^*) / \text{Downside Deviation}$$

$$17. \text{Gain-to-loss Ratio} = \left(\frac{\text{No of months with +ve R}}{\text{No of months with -ve R}} \right) \times \left(\frac{\text{Avg up month R}}{\text{Avg down month R}} \right)$$

$$18. \text{Calmar ratio} = \text{Compound Annualized ROR} / \text{ABS}^* (\text{Maximum Drawdown})$$

$$19. \text{Sterling ratio} = \text{Compound Annualized ROR} / \text{ABS}^* (\text{Average Drawdown} - 10\%)$$

where, *ABS = Absolute Value

Reading 25: Risk Management

1. Delta Normal Method: $\text{VAR} = E(R) - z\text{-value (S.D)}$
 - Daily $E(R) = \text{Annual } E(R) / 250$

- Daily S.D = Annual S.D. / $\sqrt{250}$
- Monthly $E(R) = \text{Annual } E(R) / 12$
- Monthly S.D = Annual S.D. / $\sqrt{12}$
- Daily $E(R) = \text{Monthly } E(R) / 22$
- Daily S.D = Monthly S.D. / $\sqrt{22}$
- Annual VAR = Daily VAR × $\sqrt{250}$

2. Diversification effect = Sum of individual VARs – Total VAR
3. Incremental VAR = Portf's VAR inclu a specified asset – Portf's VAR exclu that asset.
4. Tail Value at Risk (TVAR) or Conditional Tail Expectation = VAR + expected loss in excess of VAR
5. $\text{Value}_{\text{Long}} = \text{Spot}_t - [\text{Forward} / (1 + r)^n]$
6. $\text{Swap Value}_{\text{Long}} = \text{PV}_{\text{inflows}} - \text{PV}_{\text{outflows}}$
7. $\text{Fwd contract value}_{\text{Long}} = \left[\frac{\text{Spot Rate}_{D/F}}{(1 + R_{Fp})^{\text{Total time}}} - \frac{\text{Fwd Rate}_t}{(1 + R_{Fp})^{\text{Total time}}} \right] \times NP$
8. $\text{Sharpe Ratio} = \frac{\text{Mean portf R} - R_f}{\text{S.D of portf R}}$
9. $\text{Sortino Ratio} = \frac{\text{Mean portf R} - \text{Min acceptable R}}{\text{Downside deviation}}$
10. Risk Adjusted R on Capital = $\frac{\text{Expected R on an invst}}{\text{capital at risk measure}}$

$$11. \text{R over Max Drawdown} = \frac{\text{Expected Average R on an invst in a given yr}}{\text{max drawdown}}$$

Reading 26: Risk Management Applications of Forward and Futures Strategies

- $\beta = \text{Cov}_{SI} / \sigma_I^2$
 - Cov_{SI} = covariance b/w stock portf & index
 - σ_I^2 = var of index.
- β of stock portf = β of stock portf $\times$ MV of stock portf = $\beta_S S$
- Future $\beta = \beta_f \times f$
where, β_f = Futures contract beta
- Target level of beta exposure: $\beta_T S = \beta_S S + N_f \beta_f f$

$$N_f = \left(\frac{B_T - B_S}{B_f} \right) \left(\frac{S}{f} \right)$$

$$N_f = \frac{\text{Desired Beta Change}}{\frac{\text{Futures Beta}}{\text{Portfolio Value}}} \times \frac{\text{Portfolio Value}}{\text{Futures contract Price}}$$

*Actual futures price = Quoted futures price $\times$ Multiplier
- Reducing β to zero: $N_f = \left(\frac{-B_S}{B_f} \right) \left(\frac{S}{f} \right)$ and $\beta_T = 0$
- Effective β = Combined position R in % / Market R in %

- Synthetic Cash: Long Stock + Short Futures = Long risk-free bond
- Synthetic Stock: Long Stock = Long Rf bond + Long Futures
- Creating a Synthetic Index Fund:
 - No of futures contract = $N_f^* = \{V \times (1 + r)^T\} / (q \times f)$
where,
 N_f^* = No of futures contracts
 q = multiplier
 V = Portfolio value
 - Amount needed to invest in bonds = $V^* = (N_f^* \times q \times f) / (1 + r)^T$
 - Equity purchased = $(N_f^* \times q) / (1 + \delta)^T$
where, δ = dividend yield
 - Pay-off of N_f^* futures contracts = $N_f^* \times q \times (S_T - f)$
where, S_T = Index value at time T

Reading 27: Risk Management Applications of Options Strategies

- Covered Call = Long stock position + Short call position
 - Value at expiration = $V_T = S_T - \max(0, S_T - X)$
 - Profit = $V_T - S_0 + c_0$
 - Maximum Profit = $X - S_0 + c_0$
 - Max loss (when $S_T = 0$) = $S_0 - c_0$
 - Breakeven = $S_T^* = S_0 - c_0$
- Protective Put = Long stock position + Long Put position

- Value at expiration: $V_T = S_T + \max(0, X - S_T)$
 - Profit = $V_T - S_0 - p_0$
 - Maximum Profit = ∞
 - Maximum Loss = $S_0 + p_0 - X$
 - Breakeven = $S_T^* = S_0 + p_0$
- Bull Call Spread = Long Call (lower exercise price) + Short Call (higher exercise price)
 - Initial value = $V_0 = c_1 - c_2$
 - Value at expiration: V_T = value of long call – Value of short call = $\max(0, S_T - X_1) - \max(0, S_T - X_2)$
 - Profit = $V_T - c_1 + c_2$
 - Maximum Profit = $X_2 - X_1 - c_1 + c_2$
 - Maximum Loss = $c_1 - c_2$
 - Breakeven = $S_T^* = X_1 + c_1 - c_2$
 - Bull Put spread = Long Put (lower XP) + Short Put (higher XP). Identical to the sale of Bear Put Spread
XP = exercise price
 - Bear Put Spread = Long Put (higher XP) + Short Put (lower XP)
 - Initial value = $V_0 = p_2 - p_1$
 - Value at expiration: V_T = value of long put – value of short put = $\max(0, X_2 - S_T) - \max(0, X_1 - S_T)$
 - Profit = $V_T - p_2 + p_1$
 - Max Profit = $X_2 - X_1 - p_2 + p_1$
 - Max Loss = $p_2 - p_1$

f) Breakeven $= S_T^* = X_2 - p_2 + p_1$

6. Bear Call Spread = Short Call (lower XP) + Long Call (higher XP). Identical to the sale of Bull Call Spread.

7. Long Butterfly Spread (Using Call) = Long Butterfly Spread = Long Bull call spread + Short Bull call spread (or Long Bear call spread)

Long Butterfly Spread = (Buy the call with XP of X_1 and sell the call with XP of X_2) + (Buy the call with XP of X_3 and sell the call with XP of X_2).

where, $X_1 < X_2 < X_3$ and Cost of X_1 (c_1) > Cost of X_2 (c_2) > Cost of X_3 (c_3)

- a) Value at expiration: $V_T = \max(0, S_T - X_1) - 2 \max(0, S_T - X_2) + \max(0, S_T - X_3)$
- b) Profit $= V_T - c_1 + 2c_2 - c_3$
- c) Max Profit $= X_2 - X_1 - c_1 + 2c_2 - c_3$
- d) Maximum Loss $= c_1 - 2c_2 + c_3$
- e) Two breakeven points
 - i. Breakeven $= S_T^* = X_1 + \text{net premium} = X_1 + c_1 - 2c_2 + c_3$
 - ii. Breakeven $= S_T^* = 2X_2 - X_1 - \text{Net premium} = 2X_2 - X_1 - (c_1 - 2c_2 + c_3) = 2X_2 - X_1 - c_1 + 2c_2 - c_3$

8. Short Butterfly Spread (Using Call) = Selling calls with XP of X_1 and X_3 and buying two calls with XP of X_2 .

- Max Profit $= c_1 + c_3 - 2c_2$

9. Long Butterfly Spread (Using Puts) = (Buy put with XP of X_3 and sell put with XP of X_2) + (Buy the put with XP of X_1 and sell the put with XP of X_2)
where, $X_1 < X_2 < X_3$ and Cost of X_1 (p_1) < Cost of X_2 (p_2) < Cost of X_3 (p_3)

10. Short Butterfly Spread (Using Puts) = Short butterfly spread = Selling puts with XPs of X_1 and X_3 and buying two puts with XP of X_2 .

- Max Profit $= p_3 + p_1 - 2p_2$

11. For zero-cost collar

- a) Initial value of position $= V_0 = S_0$
- b) Value at expiration: $V_T = S_T + \max(0, X_1 - S_T) - \max(0, S_T - X_2)$
- c) Profit $= V_T - V_0 = V_T - S_0$
- d) Max Profit $= X_2 - S_0$
- e) Max Loss $= S_0 - X_1$
- f) Breakeven $= S_T^* = S_0$

12. Straddle = Buying a put and a call with same strike price on the same underlying with the same expiration; both options are at-the-money.

- a) Value at expiration: $V_T = \max(0, S_T - X) + \max(0, X - S_T)$
- b) Profit $= V_T - p_0 - c_0$
- c) Max Profit $= \infty$
- d) Max Loss $= p_0 + c_0$
- e) Breakeven $= S_T^* = X \pm (p_0 + c_0)$

13. Short Straddle: Selling a put and a call with same strike price on the same underlying with the same expiration; both options are at-the-money.

- Adding call option to a straddle "Strap".
- Adding put option to a straddle "Strip".

14. Long Strangle = buying the put and call on the same underlying with the same expiration but with *different exercise prices*.

15. Short Strangle = selling the put and call on the same underlying with the same expiration but with *different exercise prices*.

16. Box-spread = Bull spread + Bear spread

17. Long Box-spread = (buy call with XP of X_1 and sell call with XP of X_2) + (buy put with XP of X_2 and sell put with XP of X_1).

- a) Initial value of the box spread = Net premium $= c_1 - c_2 + p_2 - p_1$.
- b) Value at expiration: $V_T = X_2 - X_1$
- c) Profit $= X_2 - X_1 - (c_1 - c_2 + p_2 - p_1)$
- d) Max Profit = same as profit
- e) Max Loss = no loss is possible given fair option prices

- f) Breakeven = S_T^* = no break-even; the transaction always earns R_f rate, given fair option prices.

18. Pay-off of an interest rate Call Option =
 $(NP) \times \max(0, \text{Underlying rate at expiration} - X\text{-rate}) \times \left(\frac{\text{Days in underlying rate}}{360}\right)$

19. Pay-off of an interest rate Put Option =
 $(NP) \times \max(0, X\text{-rate} - \text{Underlying rate at expiration}) \times \left(\frac{\text{Days in underlying rate}}{360}\right)$

20. Loan Interest payment = $NP \times (\text{LIBOR on previous reset date} + \text{Spread}) \times \left(\frac{\text{Days in settlement period}}{360}\right)$

21. Cap Pay-Off = $NP \times (0, \text{LIBOR on previous reset date} - X \text{ rate}) \times \left(\frac{\text{Days in settlement period}}{360}\right)$

22. Floorlet Pay-Off = $NP \times (0, X \text{ rate} - \text{LIBOR on previous reset date}) \times \left(\frac{\text{Days in settlement period}}{360}\right)$

23. Effective Interest = Interest received on the loan + Floorlet pay-off

24. $\Delta = \frac{\text{Change in Option Price}}{\text{Change in Underlying Price}} = \frac{\Delta C}{\Delta S}$

25. Size of the Long position = $N_c / N_s = -1 / (\Delta C / \Delta S) = -1 / \Delta$

where, N_c = No of call options
 N_s = No of stocks

26. Hedging using non-identical option:

- One option has a delta of Δ_1 .
- Other option has a delta of Δ_2 .
- Value of the position = $V = N_1 c_1 + N_2 c_2$
 where, N = option quantity & c = option price
- To delta hedge: Desired Quantity of option 1 relative to option 2 = $\frac{\text{Delta of option 2}}{\text{Delta of option 1}}$
 $N_1 / N_2 = -\Delta c_2 / \Delta c_1$

27. $\Gamma = \frac{\text{Change in delta}}{\text{Change in underlying price}}$

28. Gamma hedge = Position in underlying + Positions in two options

29. $\text{Vega} = \frac{\text{Change in Option price}}{\text{Change in Volatility of the underlying}}$

Reading 28: Risk Management Applications of Swap Strategies

- NP of a swap (to manage D of portf.) =
 $NP = V_P \left(\frac{MD_{\text{target}} - MD_B}{MD_{\text{swap}}} \right)$
- Inverse Floater Coupon rate = $b - \text{LIBOR}$
- When $\text{LIBOR} > b$, inverse floater issuer should buy an interest rate cap with the following features:

- exercise rate of b
- $NP = NP$ of inverse floater
- Each caplet expires on the interest rate reset date of the swap/loan
- Whenever $\text{Libor} > b$, Caplet pay-off = $(L - b) \times NP$

4. Synthetic Dual-currency Bond = Ordinary bond issued in one currency Currency swap (with no principal payments)

Reading 29: Execution of Portfolio Decisions

- Bid-ask Spread = Ask price – Bid price
- Inside/Mkt bid-ask spread = Inside/Mkt Ask Price – Inside/Mkt Bid Price
- Mid-Quote = $\frac{\text{Mkt Bid Price} + \text{Mkt Ask Price}}{2}$
- Effective Spread = $2 \times (\text{Actual Execution Price} - \text{MidQuote})$
- Avg Effective Spread (ES) =

$$\frac{\text{ES of order 1} + \text{ES of order 2} + \dots + \text{ES of order n}}{n}$$
- Share Volume Wgtd (VW) ES = $[(V \text{ of shares traded for order 1} \times \text{ES of order 1}) + (V \text{ of shares traded for order 2} \times \text{ES of order 2}) + \dots + (V \text{ of shares traded for order n} \times \text{ES of order n})] / n$

7. VW Avg price = Avg P (security traded during the day)
Where, weight is the fraction of the day's volume associated with the trade

8. Mkt-adj Implementation Shortfall (IS) = I cost – Predicted R estimated using Mkt model

9. Trade Size relative to Available Liquidity

$$= \frac{\text{Order size}}{\text{Avg daily volume}}$$

10. Realized profit/loss = Execution price – Relevant decision price

11. Delay costs = $\frac{\text{No of shares actually traded}}{\text{Total No of shares in an order}} \times \frac{\text{Actual trading price on Day } t - \text{CP on day } t-1}{\text{Benchmark (closing) price on day } t_0}$
where CP = closing price

12. Missed Trade Opp Cost = $\frac{\text{No of shares not traded}}{\text{Total no of shares in an order}} \times \frac{\text{Cancellation price} - \text{Original B price}}{\text{Original B price}}$
where B = benchmark

13. IC = Commissions & Fees as % + Realized profit or loss + Delay costs + Missed trade opp costs

14. Estimated Implicit Costs for “Buy” = Trade Size × (Trade Price – B Price)

15. Estimated Implicit Costs for “Sale” = Trade Size × (B Price - Trade Price)

Reading 30: Monitoring and Rebalancing

- Buy and Hold Strategy:
 - Portfolio value = Investment in stocks + Floor value
 - Portfolio R = % in stock × R on stocks
 - Cushion = Investment in stocks = Portfolio value – Floor value
- Target Investment in Stocks under Constant Mix Strategy = Target proportion in stocks × Portfolio Value
- Target Investment in Stocks under Constant Proportion Strategy = Target proportion in stocks × (Portfolio Value – Floor value)

Reading 31: Evaluating Portfolio Performance

- Account's rate of return during evaluation period 't'
 - when there are no external cash flows = $\frac{\text{MV (end of period)} - \text{MV (beg of period)}}{\text{MV (beg of period)}}$
 - when a contribution received (start of the period) = $\frac{\text{MV (end of period)} - (\text{MV (beg of period)} + \text{contribution})}{\text{MV (beg of period)} + \text{Contribution}}$
 - when a withdrawal is made (start of the period) = $\frac{\text{MV (end of period)} - (\text{MV (beg of period)} - \text{contribution})}{\text{MV (beg of period)} - \text{Contribution}}$

- when a contribution is received at the end of the evaluation period = $\frac{(\text{MV (end of period)} - \text{contribution}) - \text{MV (beg of period)}}{\text{MV (beg of period)}}$
 - when a withdrawal is made at the end of the evaluation period = $\frac{(\text{MV (end of period)} + \text{contribution}) - \text{MV (beg of period)}}{\text{MV (beg of period)}}$
- TWR (when no external CFs) = $\frac{\text{Mkt value at end of period} - \text{Mkt value at beginning of period}}{\text{Mkt value at beginning of period}}$
 - TWR (entire evaluation period) = $(1 + r_{t,1}) \times (1 + r_{t,2}) \times \dots \times (1 + r_{t,n}) - 1$
 - MWR = $\text{MV}_t = \text{MV}_0(1+R)^m + \text{CF}_1(1+R)^{m-L(1)} + \dots + \text{CF}_n(1+R)^{m-L(n)}$
where,
m = No of time units in evaluation period
L(i) = No of time units by which the ith CF is separated from beg of evaluation period
 - Compound g rate or geometric mean R = $(1 + r_{t,1}) \times (1 + r_{t,2}) \times \dots \times (1 + r_{t,n})^{1/n} - 1$
Where, n = No of yrs in measurement period
 - Style = Manager's B portf - Mrkt index
 - Active Mgmt = Manager's portf – B
 - Portf R = MrkeIndex + Style + Active Mgmt

9. Periodic R on an a/c (factor-based model)

$$= \alpha p + (b_1 \times F_1) + (b_2 \times F_2) + \dots + (b_k \times F_k) + \epsilon p$$
10. Benchmark coverage =

$$\frac{\text{MV of securities that are present in both B \& portf}}{\text{Total MV of portf}}$$
11. Active position = Wght of a security in an account - Wght of the same security in B
12. Value-added R on a long-short portf =

$$\text{Portf R} - B$$
13. RoR for a long-short portf =

$$\frac{\text{P/L resulting from hedge fund strategy}}{\text{Amount of assets at risk}}$$
14.
$$\frac{\text{P/L resulting from hedge fund strategy}}{\text{Absolute value of all (long positions + short positions)}}$$
15. Fundamental rule of Active Mgmt: Impact

$$= (\text{active}) \text{ wght} \times R$$
16. Δ in value of fund = Total amount of net contributions
17. Ending value of a fund under the Net Contributions investment strategy =

$$\text{Beginning value} + \text{Net contributions}$$
18. Δ in Fund's value = End value of a fund under the Rf asset Invst strategy –

$$\text{Begvalue (i.e. ending value of the fund under the Net Contributions investment strategy)}$$
19. R-metric perspective: Incremental R contribution of the Asset Category investment strategy =
$$\sum_{i=1}^A W_i \times (R_{Ci} - R_f)$$
20. Value-metric perspective: Incremental contribution of the Asset Category investment strategy = Sum [(Each asset category's policy proportion of the Fund's beg value and all net external cash inflows)
$$\times (\text{Asset category's B RoR} - R_f \text{ rate})]$$
21. Aggregate manager B R under B level invstmnt strategy = Wghtd* Avg of IndMngr's B R
22. Return-metric perspective: Incremental return contribution of the B strategy =
$$\sum_{i=1}^A \sum_{j=1}^M W_i \times W_{ij} \times (r_{Bij} - r_{Ci})$$
23. Value-metric perspective: Incremental contribution of the B strategy = Sum [each manager's policy proportion of the total fund's beg value and net external cash inflows
$$\times (\text{manager's B R} - R \text{ of manager's asset category})]$$
24. Misfit R or Style bias = R generated by the aggregate of the managers' B - R generated by the aggregate of the asset category B
25. Return to the Investment managers level =

$$\text{Sum (active managers' returns} - \text{their benchmark returns)}$$
26. Return-metric perspective: Contribution of the Investment Managers strategy =
$$r_{IM} = \sum_{i=1}^A \times \sum_{j=1}^M W_i \times W_{ij} \times (r_{Aij} - r_{Bij})$$
27. Allocation Effects incremental contribution = Fund's ending value - Value calculated at the Investment Managers level
28. Value-added/active return =
$$\text{Portf R} - B R$$
29. Security-by-security analysis:
$$r_i = \sum_{i=1}^n [(W_{pi} - W_{Bi}) \times (r_i - r_B)]$$
30. Value-added return under Holdings-based or "buy-and-hold" attribution =
$$\sum_{j=1}^S W_{pj} \times r_{pj} - \sum_{j=1}^S W_{Bj} \times r_{Bj}$$
31. Value-added Return = Pure sector allocation + Allocation/selection interaction + Within sector selection
32. Pure sector Allocation =
$$\sum_{i=1}^n (W_{pj} - W_{Bj}) \times (r_{Bj} - r_B)$$
33. Within sector Selection =
$$\sum_{j=1}^S W_{Bj} (r_{pj} - r_{Bj})$$
34. Allocation/selection Interaction =
$$\sum_{i=1}^n (W_{pj} - W_{Bj}) \times (r_{pi} - r_{Bj})$$

35. Interest rate Mgmt contribution = Agg
R(re-priced securities) - R of entire
Treasury universe
36. Sector/quality return = Gross R - External
interest rate effect - Interest rate Mgmt
effect
37. Security selection effect for each security =
Total R of a security - all the other
components.
38. Portf security selection effect = Mkt value
WghtdAvg of all individual security
selection effects
39. Trading activity = Total Portf R – (Interest
rate mgmt effect + sector/quality effect +
security selection effect)
40. Alpha = $\alpha = r_P - [r_f + \beta_P(r_M - r_f)]$
41. Treynor's measure = $T_A = \frac{\bar{R}_A - \bar{r}_f}{\hat{\beta}_A}$
42. Sharpe ratio = $\frac{\bar{R}_A - \bar{r}_f}{\hat{\sigma}_A}$
43. $M^2 = \bar{r}_f + \left[\frac{\bar{R}_A - \bar{r}_f}{\hat{\sigma}_A} \right] \hat{\sigma}_M$
44. Information ratio = $IR_A = \frac{\bar{R}_A - \bar{R}_B}{\hat{\sigma}_{A-B}}$